4.2 Reflections

Name _____ Period ____ Date _____

Geometry 4.2 · Student sheet

I can…

- reflect a figure in the x-axis, the y-axis, or a horizontal or vertical line.
- reflect a figure in y = x or y = −x.
- perform a glide reflection and find lines of symmetry.

Vocabulary · fill in the missing word

Reflection — a _____ that uses a line like a mirror to reflect a figure

Line of reflection — the line that acts as the _____

Glide reflection — a _____ followed by a reflection in a line parallel to the direction of the translation

Line symmetry — a figure has line symmetry when a _____ maps the figure onto itself

Part 1 · Discovery: fold and flip

Plot A(1, 2), B(4, 5), C(5, 1). Fold along each axis and record the images.

Point	Reflect in the x-axis	Reflect in the y-axis
A(1, 2)		
B(4, 5)		
C(5, 1)		

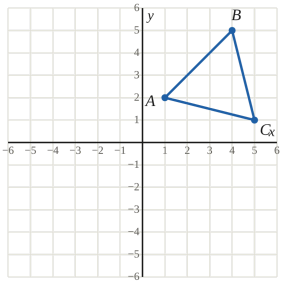
x-axis: (x, y) → (____ , ____) y-axis: (x, y) → (____ , ____)

Part 2 · Notes and examples (fill in each note, then work the example)

Reflections in the axes: in the x-axis, (x, y) → _____; in the y-axis, (x, y) → _____.

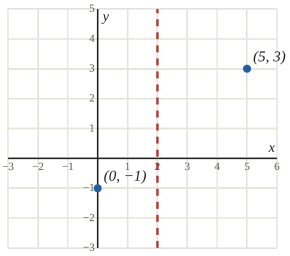
Example 1

Reflect △ABC in the x-axis, then (separately) in the y-axis. Give the vertices.



Example 2

Reflect (5, 3) and (0, −1) in the line x = 2.



Reflections in y = x and y = −x: in y = x, (x, y) → _____; in y = −x, (x, y) → _____.

Example 3

Reflect (3, −4) in the line y = x, then (separately) in the line y = −x.

Glide reflection: a translation followed by a _____ in a line parallel to the direction of the translation.

Example 4

Translate A(−3, 2) by (x, y) → (x + 4, y), then reflect in the x-axis. Find the final image.

Line symmetry: a figure has line symmetry when a _____ maps the figure onto itself.

Example 5

How many lines of symmetry does each figure have?

a. a rectangle that is not a square

b. an equilateral triangle

c. a regular hexagon

Part 3 · Partner practice

Partner 1

Reflect (−2, 5) in the x-axis and (separately) in the y-axis.

Partner 2

Reflect (6, 1) in the line y = x.